

Microsoft Teams Voice Implementation Plan

CUCM to Microsoft Teams Voice - Cloud Migration | Sandrina Sample Voice Implementation plan

Document Purpose

This implementation plan provides a structured, step-by-step approach for a business environment that currently uses Cisco Unified Communications Manager (CUCM) and wants to transition to Microsoft Teams Voice as a cloud-based communications solution. It is written as a portfolio-ready sample plan and assumes a controlled enterprise migration with coexistence, pilot validation, phased cutover, and rollback planning.

1. Initiate the Program

- Define business goals, success measures, scope, budget, stakeholders, and target completion dates.
- Identify executive sponsor, project manager, unified communications lead, Microsoft 365/Teams administrator, network team, security team, service desk, carrier/provider, and business representatives.
- Establish governance, change control, communications, escalation paths, and documentation standards.

2. Assess the Existing CUCM Environment

- Inventory CUCM clusters, nodes, versions, gateways, trunks, dial peers, route patterns, translation patterns, partitions, calling search spaces, voicemail, contact center components, devices, endpoints, and applications.
- Export current dial plan, extension assignments, DIDs, hunt groups, auto attendants/IVR functions, call queues, emergency-calling configuration, and service numbers.
- Document carrier connections, PSTN services, SIP/PRI/TDM circuits, SBCs, public IPs, DNS, firewall rules, and remote-site dependencies.
- Identify legacy phones, analog lines, elevators, alarms, fax, paging, intercoms, modems, door phones, and other non-standard endpoints.
- Assess utilization, call volume, critical call flows, after-hours routing, and business continuity requirements.

3. Define the Target Teams Voice Architecture

- Select the Teams Phone architecture and determine whether the organization will use Microsoft Calling Plans, Operator Connect, or Direct Routing based on geographic, regulatory, carrier, feature, and integration requirements.
- For Direct Routing, design a supported session border controller (SBC) architecture, SIP connectivity, certificate model, firewall policy, routing, redundancy, and carrier interconnection.
- Define Microsoft 365 tenant dependencies, licensing, identity synchronization, authentication, and administrative roles.
- Design number management, normalization, emergency calling, locations, network/site definitions, and administrative policies.
- Create high-level and detailed solution diagrams showing user, Teams, SBC/carrier, PSTN, network, and management relationships.

4. Prepare Microsoft 365 and Teams

- Confirm tenant readiness and required Microsoft Teams Phone licensing.
- Create or validate administrative role assignments and least-privilege access.
- Configure voice-related policies, emergency locations, calling settings, resource accounts, auto attendants, and call queues as applicable.
- Establish naming standards, user provisioning process, number-assignment workflow, and support procedures.

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- Prepare reporting, monitoring, audit, and operational dashboards.

5. Prepare Network and Security

- Validate WAN/internet capacity, latency, jitter, packet loss, DNS, NTP, routing, and QoS readiness for real-time voice.
- Review firewall, proxy, NAT, SBC, TLS, SIP, and media-flow requirements.
- Define trusted administration paths and security controls for Teams, SBCs, carrier connectivity, and privileged accounts.
- Confirm emergency calling location accuracy and network-to-location mapping.
- Document contingency paths for sites with unreliable connectivity.

6. Build the Coexistence Design

- Define how CUCM and Teams users will coexist during the migration period.
- Create temporary call-routing paths between CUCM and Teams.
- Determine migration sequencing for DIDs, extensions, hunt groups, service numbers, and shared call flows.
- Identify users and services that must remain on CUCM until dependencies are removed.
- Document rollback routing so calls can be returned to CUCM if a cutover issue occurs.

7. Configure the Pilot

- Select a representative pilot group across user types, locations, devices, and call patterns.
- Assign licenses, enable Teams Phone, provision numbers, configure emergency location, and apply required policies.
- Configure a limited set of business call flows and service numbers for pilot validation.
- Complete endpoint readiness checks for Teams desktop, web, mobile, desk phones, headsets, conference devices, and shared devices as applicable.

8. Pilot Validation and Defect Resolution

- Execute functional, integration, network/QoS, emergency-calling, security, resiliency, and user-acceptance tests.
- Capture test evidence, defects, severity, owner, corrective action, retest result, and closure approval.
- Resolve critical and high-impact issues before expanding the migration.
- Confirm operational support procedures and knowledge transfer.

9. Prepare the Production Migration

- Freeze or control changes to the affected CUCM dial plan before the migration window.
- Finalize migration waves by location, department, user type, and business criticality.
- Validate number porting or carrier activation dates and dependencies.
- Confirm support staffing, bridge calls, communications, escalation contacts, and executive notifications.
- Publish a detailed cutover runbook with task owners and timestamps.

10. Execute Migration Waves

- Complete pre-cutover checks and confirm entry criteria.
- Move users and numbers in controlled waves.

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- Validate registration, inbound/outbound calling, internal dialing, transfers, conferencing, voicemail, call queues, auto attendants, and emergency calling.
- Monitor call quality, SBC/carrier health, incidents, and user reports after each wave.
- Record exit criteria and obtain approval before proceeding to the next wave.

11. Decommission CUCM Dependencies

- Identify all remaining CUCM integrations and services.
- Retire unused devices, trunks, gateways, dial-plan objects, licenses, and administrative dependencies in accordance with change-control policy.
- Preserve required configuration records and support documentation.
- Validate that business services continue to function after each decommissioning step.

12. Operational Handover

- Deliver as-built architecture, configuration standards, test results, known issues, support procedures, escalation matrix, and monitoring procedures.
- Train service desk and operations personnel on Teams Voice troubleshooting and user support.
- Establish ongoing governance for number management, emergency locations, policy changes, carrier management, and service reviews.

Implementation Sequence

Phase	Primary Activities	Exit Criteria
1. Initiation	Charter, scope, governance, stakeholders	Approved project charter
2. Discovery	CUCM, carrier, network, endpoints, call flows	Validated inventory and dependency map
3. Architecture	Teams Voice, SBC/carrier, security, numbering	Approved target design
4. Readiness	Tenant, licensing, network, security	Readiness checklist passed
5. Pilot	Pilot users, configuration, testing	Pilot exit criteria met
6. Migration	Wave cutovers, validation, monitoring	Wave acceptance achieved
7. Decommission	Remove legacy dependencies	CUCM retirement criteria met
8. Handover	Documentation, training, operations	Operational acceptance

Key Cutover Entry Criteria

- Target architecture approved and implemented.
- Carrier/numbering readiness confirmed.
- Emergency calling validated.
- Pilot defects at acceptable status.
- Rollback plan tested and documented.

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- Business and technical support coverage confirmed.

Rollback Approach

If defined cutover exit criteria are not met, stop the migration wave, restore documented CUCM routing and affected service paths, confirm inbound/outbound calling, notify stakeholders, capture the incident and root cause, and reschedule only after corrective actions are validated.

Portfolio Disclaimer

This document is a sample implementation plan created for demonstration and portfolio purposes. It uses generalized business and technical scenarios and is not a reproduction of proprietary employer or customer documentation.